

ME 331

Lab 1: Pin Fin

Objective

The purpose of this experiment is to determine the convection heat transfer coefficients, h , for four different pin fins operating at steady state. Heat transfer is occurring in each fin via conduction and free convection.

Background

Heat transfer theory provides a direct means of developing engineering estimates of the rate of heat transfer in various practical situations. For convective heat transfer, empirical correlations are developed with a theoretical basis, but with experimentally determined parameters. Such empirical correlations can represent idealizations that are not typically realized, even in the laboratory. It is often not appreciated that convection correlations are seldom better than 20% accurate, and can frequently be even worse. Thus, although the correlations can be used for engineering estimates, more exact information is obtained by the construction and testing of a prototype. In this experiment, the temperature distributions for four different pin fins with natural convections are measured: a copper square, a copper round, a stainless-steel rod and aluminum rod.

Experimental Setup

The experimental setup consists of four different pin fin configurations. The heater and base of each fin is surrounded by insulation to prevent heat loss. Mounted along the length of the fins are 4 to 6 thermocouples. Using the data acquisition system, thermocouple temperatures are displayed and recorded. The heaters are powered from a VARIAC, so the power supplied to each fin varies. The knob on each fin allows the power input to each heater to be adjusted. Two Keithly meters measure the voltage and current of the heaters. The switchbox is used to set which fin is being monitored by the Keithly meters.

The appendices of this handout contain helpful drawings of the setup, in addition to important information about the geometry and material properties of the fins.

Procedure

Note that the fins will already be at steady state when you arrive. There are two stations, each with two fins. At each station, you can only collect data from one fin at a time. You will collect temperature data for four of the fins under steady state, free convection conditions: a copper square, a copper round, a stainless-steel rod and aluminum rod.

1. Starting at one of the two stations, plug in the USB cord for one of the fins, noting the fin geometry and material. Make sure to only plug in one USB cord at a time.
2. Open the ME331 folder on the desktop.
3. Open the file Fin Experiment.py. This will open two windows: a Python shell that will show the program outputs, and the program code itself.
4. In the Fin Experiment.py window, select Run > Run Module, or press F5. It may take a minute for the program to load.
5. Once the program starts, press 'Start' to begin taking data and plotting. Take data for 10-30 seconds at steady state.
6. After you have finished collecting data, press 'Save' and save your file in the ME331 folder on the desktop, using the group name as the file name.
7. To switch to a different fin, close the window with the plots, unplug the USB cord of the previous fin, plug in the new fin, and start again at step 4. Make sure to note the geometry and material for each fin.
8. Write down the current and voltage supplied to each fin.
9. Transfer your data to a flash drive, switch stations, and repeat the procedure to collect the remaining data.

Analysis

Theory

Consider a fin composed of an isotropic material of uniform cross-sectional area operating at steady state with no internal energy generation. Assuming one-dimensional conduction, a fixed temperature at the base of the fin, and convection from the tip of the fin, the temperature distribution is given as:

Equation 1:
$$\frac{\theta(x)}{\theta_b} = \frac{\cosh m(L-x) + (h/mk) \sinh m(L-x)}{\cosh mL + (h/mk) \sinh mL}$$

Where:

- $\theta(x) = T(x) - T_\infty$
- $\theta_b = T_b - T_\infty$
- $m = \sqrt{hP/kA_c}$

Variable Definitions: Note that $T(x)$ is the temperature distribution in the x direction, T_∞ is the ambient temperature, T_b is the temperature at the base of the fin, h is the convection coefficient, P is the perimeter of the fin, k is the thermal conductivity, A_c is the cross-sectional area of the fin, and L is the length of the fin.

The rate of heat transfer from the fin is given as:

Equation 2: $q_f = M \frac{\sinh mL + (h/mk) \cosh mL}{\cosh mL + (h/mk) \sinh mL}$

Where: $M = \theta_b \sqrt{hPkA_c}$

Calculations

1. Using the measured temperature distribution and equation (1), calculate the convection coefficient h for the square copper fin, cylindrical copper fin, cylindrical stainless steel fin and aluminum rod fin. Perform this calculation by finding the value of h that minimizes S , the least squares error between the measured temperature distribution and the analytical temperature distribution. The least squares error S is given as:

$$S = \sum_{i=1}^N \left[\left(\frac{\theta}{\theta_b} \right)_{measured} - \left(\frac{\theta}{\theta_b} \right)_{analytical} \right]^2$$

where N is the number of measurements (in this case, the number of thermocouples).

2. Using the computed h values, calculate the heat transfer rate q_f from each fin using equation (2).

Results

In the results section, include your data plotted as temperature as a function of x position for each fin. Include a table that presents the measured rate of heat transfer into each fin from the heaters. Describe your data analysis methods and present the results of your calculations.

Discussion

Address the following discussion points in your lab report. You may choose to include additional discussion.

1. Discuss the reasons for the variation in temperature between the base and the tip thermocouples. Why is the difference between the base and tip thermocouples larger for some fins than others? How do the geometry and material properties contribute to this?
2. How do the geometry and material properties contribute to the difference in convection heat transfer rates between fins?
3. Compare the calculated heat transfer rate q_f to the rate of heat transferred into the fin from the heater. According to theory, should these values be equal? Why? If they are not equal, what factors could have contributed to the difference?

Appendices

Appendix A: Fin Properties and Thermocouple Locations

Note: All thermocouples are inserted 0.64 cm deep into the fins.

Table 1: Fin Materials and Dimensions

Fin Material	Dimensions (cm)	Thermal Conductivity ($W/m \cdot K$)
Copper Alloy 110 (square)	1.27 W $\times$ 1.27 H $\times$ 28.50 L	388
Copper Alloy 110 (round)	1.27 D $\times$ 17.20 L	388
Stainless Steel Alloy 303 (round)	0.95 D $\times$ 28.50 L	16
Aluminum Alloy 6061-T6 (round)	1.27 D $\times$ 28.50 L	167

Table 2: Thermocouple Distances from Base

Thermocouple #	Stainless Rod (cm)	Copper Rod (cm)	Copper Square (cm)	Aluminum Rod (cm)
1	3.10	3.09	3.06	4.35
2	5.44	5.56	5.50	5.41
3	9.81	9.80	9.82	9.72
4	17.22	17.15	17.15	17.15
5	22.07	N/A	22.07	22.07
6	28.45	N/A	28.07	28.07

Appendix B: Operating Conditions

Heater voltage settings are chosen to limit the base temperature to a maximum of 100°C.

Table 3: Maximum Heater Settings

Fin Material	Max Voltage Setting (V)	Base Temperature (°C)	Current (A)	Power (W)
Stainless Steel	35.00	87	0.182	6.37
Copper, Square	40.00	68	0.216	8.64
Copper, Round	40.00	81	0.218	8.72
Aluminum	45.00	91	0.247	11.12

Appendix C: Experimental Setup Drawing

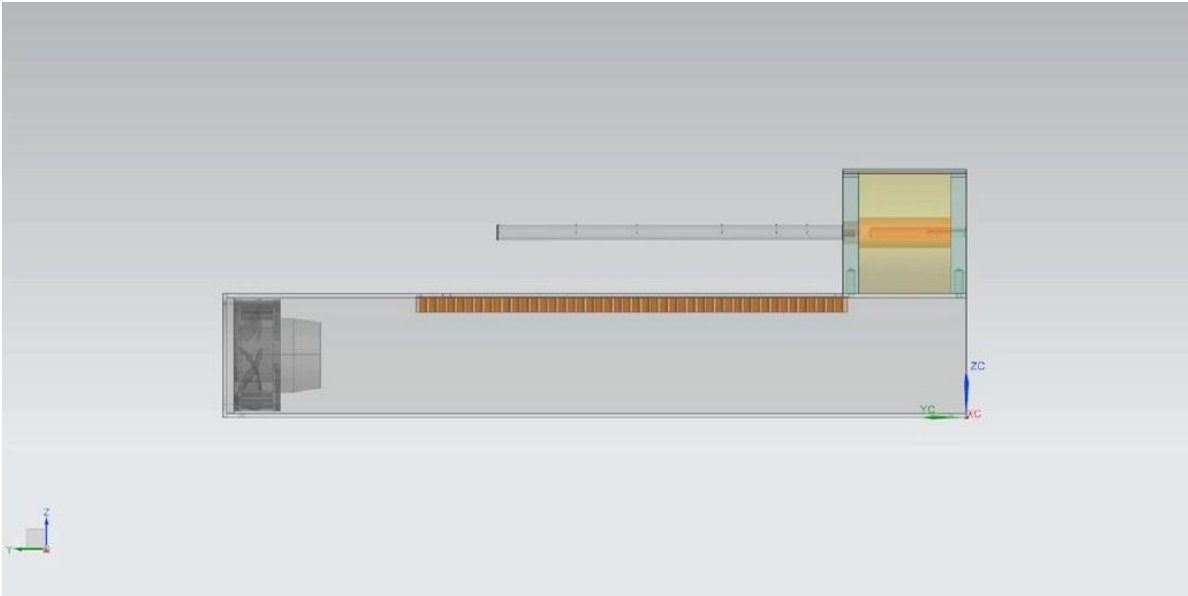


Figure 1. Fin Experiment, Side View. A 3D CAD side-profile render of the experimental apparatus box, showing the slender rod fin protruding horizontally over the workspace.]

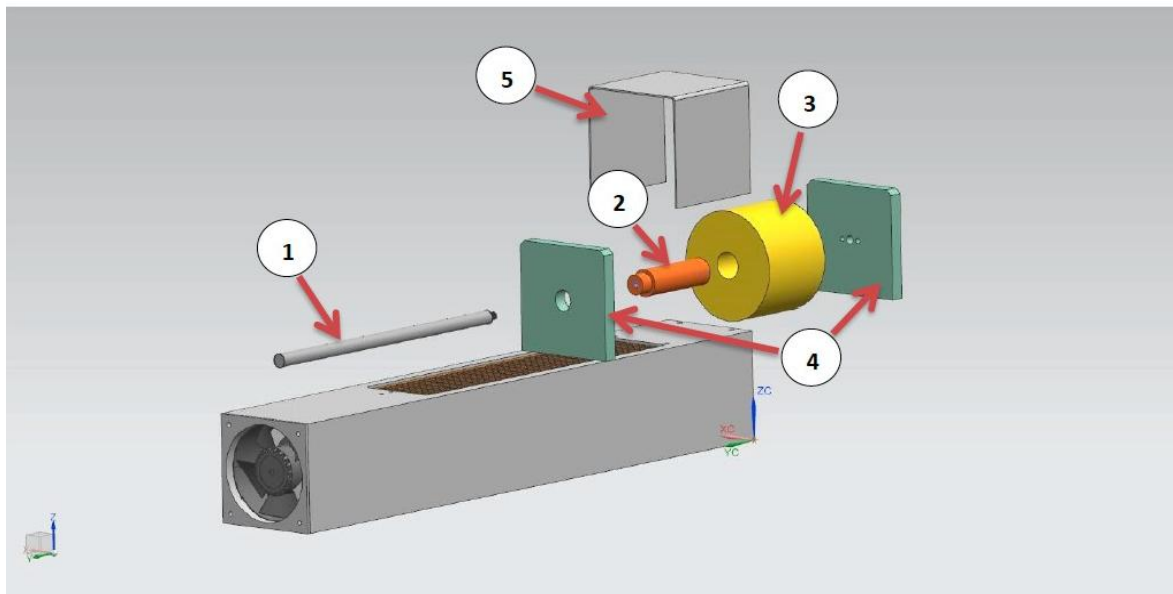


Figure 2. Fin Experiment, Exploded View. A 3D CAD exploded render showing the numbered internal components of the test station: 1. Fin, 2. Heater Jacket, 3. Insulation, 4. Heater Supports, and 5. Cover.]

Table 4: Parts List

Part	Name	Material	Description
1	Fin	Various	Slender rods with thermocouples embedded along the length
2	Heater Jacket	Copper	Cartridge heater sheath, 1" diameter x 3" long
3	Insulation	Fiberglass	Pipe insulation; 1" ID x 3" OD, K-Factor: 0.23
4	Heater Supports	Garolite (G-10)	½" x 4" x 4"; Thermal Conductivity: 0.288 W/m-K
5	Cover	Aluminum	Heater cover, alloy 3003-H14, 0.05" thick.